

Git/GitHub Introduction

Ty Bennett

What is Git?

- Created by Linus Torvalds,
 - Also creator of Linux (2005)
 - Came out of Linux development community
 - Designed to do version control on the Linux kernel
- Goals of Git:
 - Speed
 - Support for non-linear development
 - Thousands of parallel branches
 - Fully distributed
 - Able to handle large projects effectively



Why use Git?

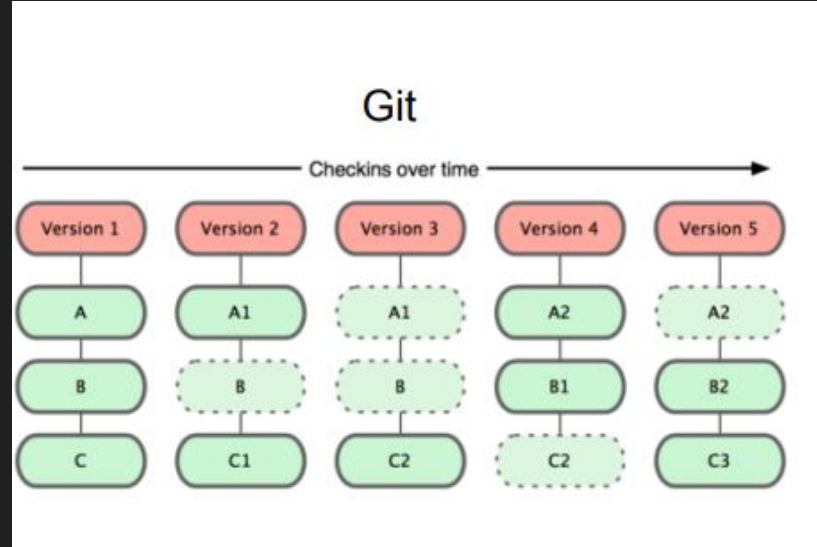
- Distributed
 - Work online or offline
- Popular and successful
 - Active development
 - Shiny new tools and ecosystems
 - Fast
- Tracks any type of file
 - From .pdf to .java to .txt
- Branching
 - Allows you to make multiple version of the same snapshot of the files
 - Smarter merges of branches

Distributed version-control system

- You keep your files in a repository on your local machine.
 - Many operations are local
- You synchronize your repository with a repository on a server.
- If your coworker uploads some changes to your files, you can pick those up by synchronizing with the server.

Snapshots

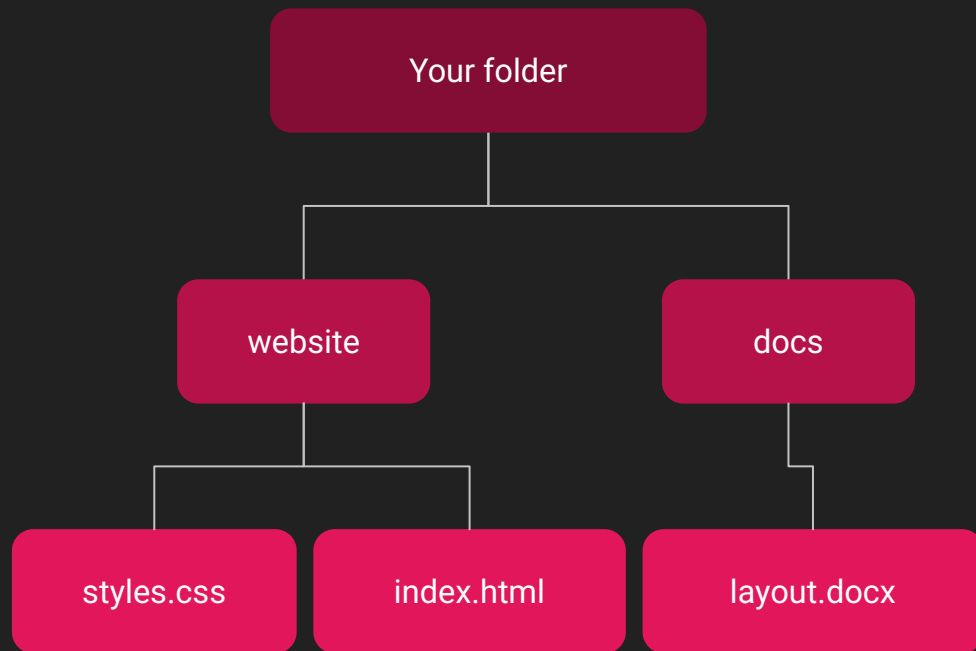
- Terminology: In git-speak, a “version” is called a “commit.”
- Git keeps “snapshots” of the entire state of the project.
- Git keeps track of the history of your commits



Installing git/Learn more

- Windows, Mac:
 - Navigate to <https://git-scm.com/downloads>
 - Follow download instructions for your OS
- Linux
 - Use your package manager to install Git
 - Package name: git
- Resources:
 - Free online book: <http://git-scm.com/book>
 - Reference page for Git: <http://gitref.org/index.html>
 - Git for Computer Scientists: <http://eagain.net/articles/git-for-computer-scientists/>
 - At command line (where verb = **config, add, commit, merge, checkout**):
 - Run *git help verb*

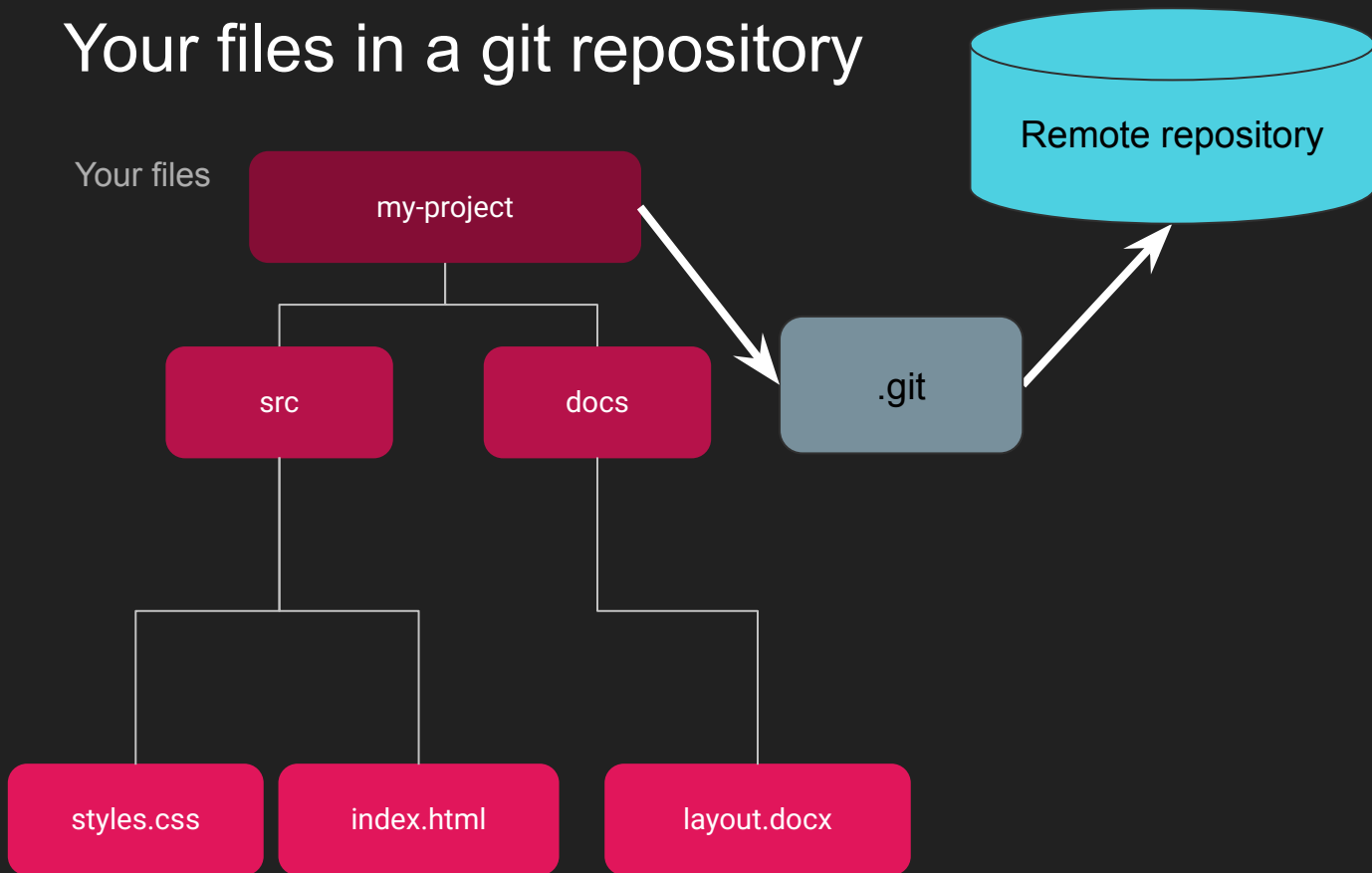
Simple model of Git



Here are your files, sitting in a directory called folder

Your files in a git repository

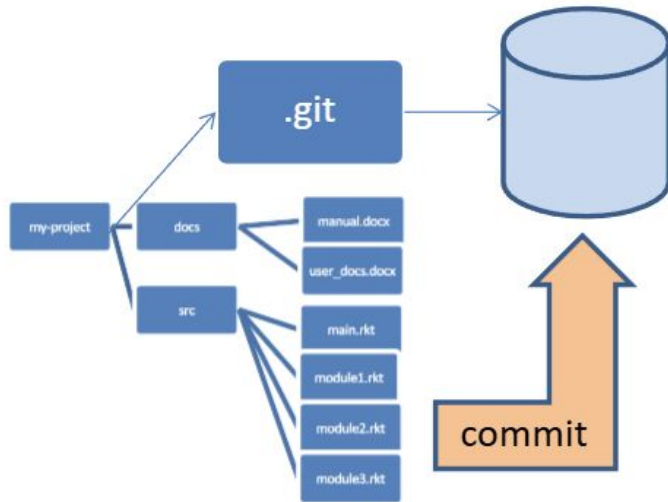
Your files



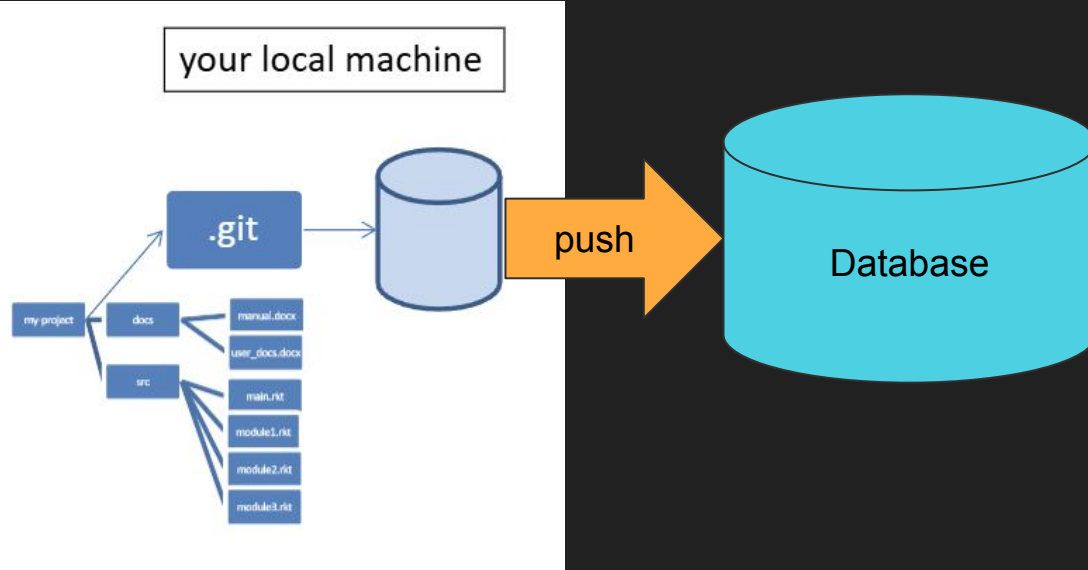
Your git workflow

- You edit your local files directly.
- You can edit, add files, delete files, etc., using whatever tools you like.
- This doesn't change the mini-filesystem, so now your mini filesystem is behind.

A commit



Synchronizing with the server



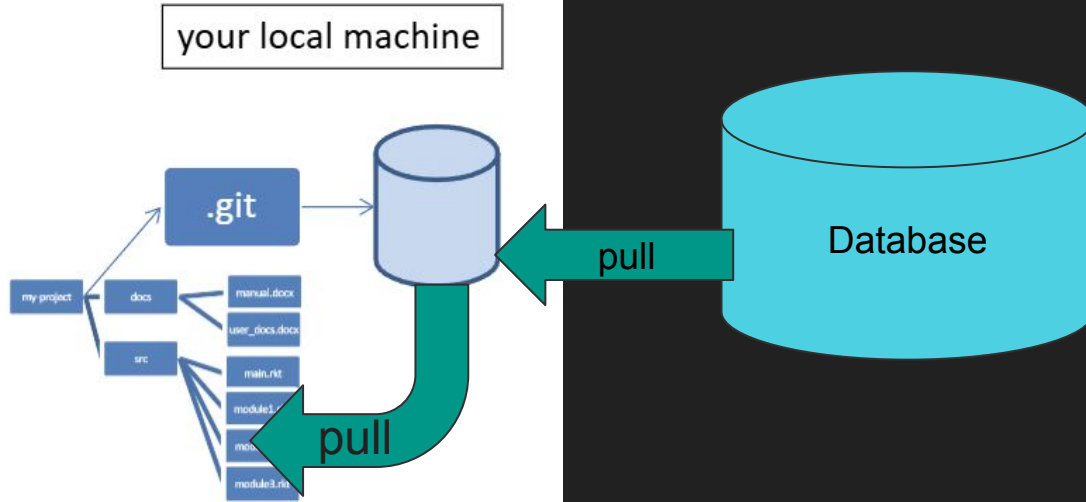
A server, somewhere on the internet
(github.com in this case)

At the end of each work session, you need to save your changes on the server. This is called a “push.”

Now all your data is backed up.

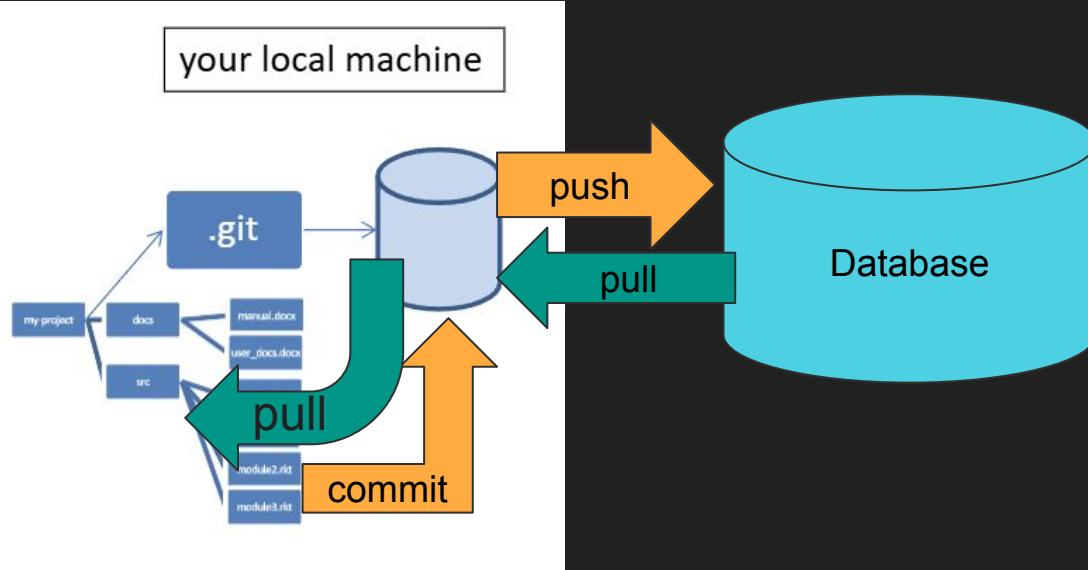
Synchronizing with the server cont.

A server, somewhere on the internet (github.com in this case)



The whole picture

A server, somewhere on the internet
(github.com in this case)



Demo time!